

BÀI TẬP TUẦN 7: Tích phân đường

1)

a) $\int_C y e^x ds$, $C: (1, 2) \rightarrow (4, 7)$

Phương trình thẳng đi qua 2 điểm trên:

d: $y = ax + b$

$$\Rightarrow \begin{cases} a + b = 2 \\ 4a + b = 7 \end{cases} \Leftrightarrow \begin{cases} a = 5/3 \\ b = 1/3 \end{cases}$$

$$\Rightarrow d: y = \frac{5x}{3} + \frac{1}{3}$$

$C: \begin{cases} x = t \\ y = \frac{5t}{3} + \frac{1}{3} \end{cases}, 1 \leq t \leq 4$

$$y = \frac{5t}{3} + \frac{1}{3}$$

$$\int_C y e^x ds = \int_1^4 \left(\frac{5}{3}t + \frac{1}{3} \right) \cdot e^t \sqrt{1^2 + \left(\frac{5}{3} \right)^2} dt$$

$$= \int_1^4 \left(\frac{5}{3}t + \frac{1}{3} \right) e^t \frac{\sqrt{34}}{3} dt$$

$$= \frac{\sqrt{34}}{3} \int_1^4 \left(\frac{5}{3}t + \frac{1}{3} \right) e^t dt \quad (*)$$

Taxét: $\int_1^4 \left(\frac{5}{3}t + \frac{1}{3} \right) e^t dt = I$

$$\text{Đặt } \begin{cases} u = \frac{5}{3}t + \frac{1}{3} \\ dv = e^t dt \end{cases} \Rightarrow \begin{cases} du = 5/3 dt \\ v = e^t \end{cases}$$

$$I = \left(\frac{5}{3}t + \frac{1}{3} \right) e^t \Big|_1^4 - \frac{5}{3} \int_1^4 e^t dt$$

$$= \left(\frac{5}{3}t + \frac{1}{3} \right) e^t \Big|_1^4 - \frac{5}{3} e^t \Big|_1^4$$

$$= 7e^4 - 2e - \frac{5}{3}(e^4 - e)$$

$$= \frac{16}{3}e^4 - \frac{e}{3}$$

$$\begin{aligned} (1) \Rightarrow \int_C ye^x ds &= \frac{\sqrt{34}}{3} \cdot \left(\frac{16}{3}e^4 - \frac{e}{3} \right) \\ &= \frac{\sqrt{34}}{9} e(16e^3 - 1) \end{aligned}$$

b)

$$\int_C xe^y ds, \quad C: x=e^t \text{ từ } (1,0) \rightarrow (e,1)$$

~~Đường dây thẳng đi qua 2 điểm~~

$$\begin{aligned} C: x &= e^t \\ y &= t, \quad 0 \leq t \leq 1 \end{aligned}$$

$$\begin{aligned} \int_C xe^y ds &= \int_0^1 e^t \cdot e^t \sqrt{(e^t)^2 + 1^2} dt \\ &= \int_0^1 e^{2t} \sqrt{e^{2t} + 1} dt \end{aligned}$$

$$\text{Đặt } u = e^{2t} + 1 \Rightarrow du = 2e^{2t} dt$$

$$\Rightarrow \int_0^1 \frac{e^{2t}}{e^{2t}+1} \sqrt{e^{2t}+1} dt$$

$$= \frac{1}{2} \int_2^{e^2+1} \sqrt{u} du$$

$$= \frac{1}{2} \cdot \frac{2u^{3/2}}{3} \Big|_2^{e^2+1}$$

$$= \frac{1}{3} u^{3/2} \Big|_2^{e^2+1} = \frac{1}{3} \left[(e^2+1)^{3/2} - 2^{3/2} \right]$$

c) $\int_C (xy + \ln x) dy$:

C: $\begin{cases} x = t \\ y = t^2 \end{cases}, 1 \leq t \leq 3$

$$\begin{aligned} & \int_C (xy + \ln x) dy \\ &= \int_1^3 (t^3 + \ln t) \sqrt{4t^2} dt \\ &= 2 \int_1^3 (t^3 + \ln t) t dt \\ &= 2 \int_1^3 t^4 + t \ln t dt \\ &= 2 \left(\int_1^3 t^4 dt + \int_1^3 t \ln t dt \right) \quad (*) \end{aligned}$$

Ta xét $\int_1^3 t \ln t dt = I$

~~Đặt $u = \ln t \Rightarrow du = \frac{1}{t} dt$~~
 $\begin{cases} u = \ln t \\ dv = t dt \end{cases} \Rightarrow \begin{cases} du = \frac{1}{t} dt \\ v = \frac{t^2}{2} \end{cases}$

$$I = \left[\frac{t^2}{2} \ln t \right]_1^3 - \frac{1}{2} \int_1^3 t dt$$

$$= \frac{9}{2} \ln 3 - \frac{1}{2} \left[\frac{t^2}{2} \right]_1^3$$

$$= \frac{9}{2} \ln 3 - \frac{1}{4} (9 - 1) = \frac{9}{2} \ln 3 - 2$$

$$(1) \Rightarrow \int_2^3 (xy + \ln x) dy$$

$$= 2 \left(\int_1^3 t^4 dt + \frac{9}{2} \ln 3 - 2 \right)$$

$$= 2 \left(\frac{t^5}{5} \Big|_1^3 + \frac{9}{2} \ln 3 - 2 \right)$$

$$= 2 \left(\frac{243}{5} - \frac{2 \cdot 33}{5} + \frac{9}{2} \ln 3 - 2 \right)$$

$$= 2 \left(42 - \frac{2 \cdot 33}{5} + \frac{9}{2} \ln 3 \right)$$

$$= \frac{464}{5} + 9 \ln 3$$

d)

$$\int_C \sin x dx + \cos y dy, \quad C \text{ nửa trên đường tròn}$$

$$x^2 + y^2 = 1$$

$$C_1: \begin{cases} x = t \\ y = \sqrt{1 - t^2} \end{cases}, \quad -1 \leq t \leq 1$$

$$C_2: \text{pt đoạn thẳng đi qua } (-1, 0), (-2, 3)$$

$$C_2: y = ax + b$$

$$\Rightarrow \begin{cases} -a + b = 0 \\ -2a + b = 3 \end{cases} \Leftrightarrow \begin{cases} a = -3 \\ b = -3 \end{cases}$$

$$\Rightarrow C_2: y = -3x - 3$$

$$\Rightarrow C_2: \begin{cases} x = t \\ y = -3t - 3 \end{cases}, \quad -2 \leq t \leq -1$$

• ~~th~~

$$\int_C \sin x dx + \cos y dy = \int_{C_1} \sin x dx + \cos y dy + \int_{C_2} \sin x dx + \cos y dy$$

Ta có: $\int_{C_1} \sin(x) dx + \cos(y) dy$

$$= \int_{-1}^1 \sin(t) + \cos(\sqrt{1-t^2}) \sqrt{\left(\frac{-t}{\sqrt{1-t^2}}\right)^2} dt$$

$$= \int_{-1}^1 \sin(t) dt + \int_{-1}^1 \cos(\sqrt{1-t^2}) \frac{t}{\sqrt{1-t^2}} dt \quad (*)$$

Ta xét $\int_{-1}^1 \cos(\sqrt{1-t^2}) \frac{t}{\sqrt{1-t^2}} dt = f(t)$

Đặt $f(-t) = \frac{\cos(\sqrt{1-t^2})}{\sqrt{1-t^2}} \cdot \frac{-t}{1} = -\frac{\cos(\sqrt{1-t^2})}{\sqrt{1-t^2}} t$

$$f(-t) = -\frac{t \cos(\sqrt{1-t^2})}{\sqrt{1-t^2}} = -f(t)$$

$\Rightarrow f(t)$ là hàm lẻ $\Rightarrow \int_{-1}^1 f(t) dt = 0$

(1) $\Rightarrow \int_{C_1} \sin(x) dx + \sin(y) dy$

$$= \int_{-1}^1 \sin(t) dt + 0 = -\cos(t) \Big|_{-1}^1 \rightarrow \cos$$

$$= -\cos(1) + \cos(-1)$$

$$= 0$$

Ta có:

$$\int_{C_2} \sin(x) dx + \cos(y) dy$$

$$= \int_{-2}^{-1} \sin(t) dt + \cos(-3t-3) \sqrt{(-3)^2} dt$$

$$= \int_{-2}^{-1} \sin(t) dt + 3 \int_{-2}^{-1} \cos(-3t-3) dt$$

$$= \int_{-2}^{-1} \sin(t) + 3 \cos(-3t-3) dt$$

$$= \sin(-t) - \cos(t) - \frac{3}{3} \sin(-3t-3) \Big|_{-2}^{-1}$$

$$= -\cos(-1) - \sin(0) - [-\cos(-2) - \sin(3)]$$

$$= -\cos(1) + \cos(2) + \sin(3)$$

$$\int_a^b \sin(x) dx + \cos(y) dy$$

$$= \int_{C_1} \sin(x) dx + \cos(y) dy + \int_{C_2} \sin(x) dx + \cos(y) dy$$

$$= 0 + -\cos(1) + \cos(2) - 3\sin(3)$$

$$= -\cos(1) + \cos(2) - 3\sin(3)$$

e)

$$\int_C xyz \, ds, \quad C: x = 2\sin(t), y = t, z = -2\cos(t) \\ 0 \leq t \leq \pi$$

$$\int_C xyz \, ds$$

$$= \int_0^\pi 2\sin(t) \cdot t \cdot (-2\cos(t)) \cdot \sqrt{(2\cos(t))^2 + 1^2 + (2\sin(t))^2} dt$$

$$= -4 \int_0^\pi \sin(2t) t \cdot \sqrt{5} dt$$

$$= -2\sqrt{5} \int_0^\pi \sin(2t) t \, dt \quad (*)$$

$$\text{Đặt } \begin{cases} u = t \\ dv = \sin(2t) dt \end{cases} \Rightarrow \begin{cases} du = dt \\ v = -\frac{1}{2} \cos(2t) \end{cases}$$

$$\begin{aligned}
 (1) &\Rightarrow -2\sqrt{5} \int_0^{\pi} \sin(2t) \, dt \\
 &= -2\sqrt{5} \left(-\frac{t}{2} \cos(2t) \Big|_0^{\pi} + \frac{1}{2} \int_0^{\pi} \cos(2t) \, dt \right) \\
 &= -2\sqrt{5} \left(-\frac{\pi}{2} + \frac{1}{2} \sin(2t) \Big|_0^{\pi} \right) \\
 &= -2\sqrt{5} \left(-\frac{\pi}{2} \right) = \sqrt{5} \pi
 \end{aligned}$$

2)

a) $F(x, y) = x^2 y^3 \vec{i} - y \sqrt{x} \vec{j}$,

$\vec{r}(t) = t^2 \vec{i} - t^3 \vec{j}$, $0 \leq t \leq 1$.

$$\begin{aligned}
 F(\vec{r}(t)) &= t^4 (-t^3) \vec{i} - (-t)^3 \sqrt{t^2} \vec{j} \\
 &= -t^{13} \vec{i} + t^4 \vec{j}
 \end{aligned}$$

$\vec{r}'(t) = \langle 2t, -3t^2 \rangle$

$\int_C \vec{F} \, d\vec{r}$

$$= \int_0^1 (-t^{13}) \cdot 2t + t^4 \cdot (-3)t^2 \, dt$$

$$= \int_0^1 -2t^{14} - 3t^6 \, dt$$

$$= \left[-\frac{2t^{15}}{15} - \frac{3t^7}{7} \right]_0^1$$

$$= -\frac{2}{15} - \frac{3}{7} = -\frac{14}{105} - \frac{45}{105} = -\frac{59}{105}$$

$$b) F(x, y, z) = yz \vec{i} + xz \vec{j} + xy \vec{k}$$

$$\vec{r}(t) = t \vec{i} + t^2 \vec{j} + t^3 \vec{k}, \quad 0 \leq t \leq 2$$

$$\vec{r}'(t) = \langle 1, 2t, 3t^2 \rangle$$

$$F(\vec{r}(t)) = t^5 \vec{i} + t^4 \vec{j} + t^3 \vec{k}$$

$$\int_C \vec{F} d\vec{r} = \int_0^2 t^5 + 2t^5 + 3t^3$$

$$= \int_0^2 t^5 + 2t \cdot t^4 + 3t^2 \cdot t^3 dt$$

$$= \int_0^2 t^5 + 2t^5 + 3t^5 dt$$

$$= 6 \int_0^2 t^5 dt$$

$$= \left. \frac{6t^6}{6} \right|_0^2 = t^6 \Big|_0^2 = 2^6 = 64$$

$$c) F(x, y) = e^{x-1} \vec{i} + xy \vec{j}$$

$$\vec{r}(t) = t^2 \vec{i} + t^3 \vec{j}, \quad 0 \leq t \leq 1$$

$$\vec{r}'(t) = \langle 2t, 3t^2 \rangle$$

$$F(\vec{r}(t)) = e^{t^2-1} \vec{i} + t^2 \cdot t^3 \vec{j}$$

$$= e^{t^2-1} \vec{i} + t^5 \vec{j}$$

$$\int_C \vec{F} d\vec{r} = \int_0^1 2e^{t^2-1} t + 3t^5 t^2 dt$$

$$= \int_0^1 2e^{t^2-1} t + 3t^7 dt$$

$$= 2 \int_0^1 e^{t^2-1} t dt + 3 \int_0^1 t^7 dt \quad (*)$$

$$\text{Ta xét: } \int_0^1 e^{t^2-1} t dt$$

Set $u = t^2 - 1 \Rightarrow du = 2t dt$

$$\Rightarrow \int_0^1 e^{t^2-1} t dt = \frac{1}{2} \int_{-1}^0 e^u du$$

$$= \frac{1}{2} e^u \Big|_{-1}^0 = \frac{1}{2} (1 - e^{-1})$$

$$W \Rightarrow \int_C \vec{F} d\vec{r} = 2 \int_0^1 \frac{1}{2} (1 - e^{-1}) + 3 \int_0^1 t^2 dt$$

$$= 1 - e^{-1} + 3 \left. \frac{t^3}{3} \right|_0^1$$

$$= 1 - e^{-1} + \frac{3}{8} = \frac{11}{8} - e^{-1}$$

d) $F(x, y, z) = z\vec{i} + y\vec{j} - x\vec{k}$

$$\vec{r}(t) = t\vec{i} + \sin t \vec{j} + \cos t \vec{k}, \quad 0 \leq t \leq \pi$$

$$\vec{r}'(t) = \langle 1, \cos t, -\sin t \rangle$$

$$F(\vec{r}(t)) = \cos t \cdot \vec{i} + \sin t \cdot \vec{j} - t \cdot \vec{k}$$

$$\int_C \vec{F} d\vec{r} = \int_0^\pi \cos t + \sin t \cdot \cos t + t \cdot \sin t dt$$

$$= \int_0^\pi \cos t + \frac{\sin 2t}{2} + t \sin t dt$$

$$= \int_0^\pi \cos t dt + \frac{1}{2} \int_0^\pi \sin 2t dt + \int_0^\pi t \sin t dt$$

Take: $I = \int_0^\pi t \sin t dt$

Set $u = t$

$dv = \sin t dt \Rightarrow \int dv = -\cos t$

$$I = -t \cos t \Big|_0^\pi + \int_0^\pi \cos t dt$$

$$= \pi + \sin t \Big|_0^\pi = \pi$$

Ta co: $\int_C \vec{F} d\vec{r} = \int_0^\pi \cos t dt + \int_0^\pi \frac{1}{2} \int_0^\pi \sin 2t dt + \int_0^\pi t \sin t dt$

$$= \sin t \Big|_0^\pi - \frac{1}{4} \cos 2t \Big|_0^\pi + \pi$$

$$= -\frac{1}{4}(1-1) + \pi = \pi$$